# Neural-Symbolic Integ
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## Survey about current and future trends

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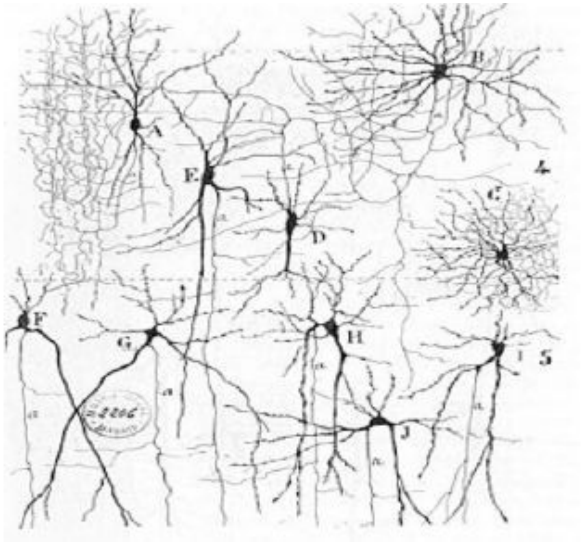
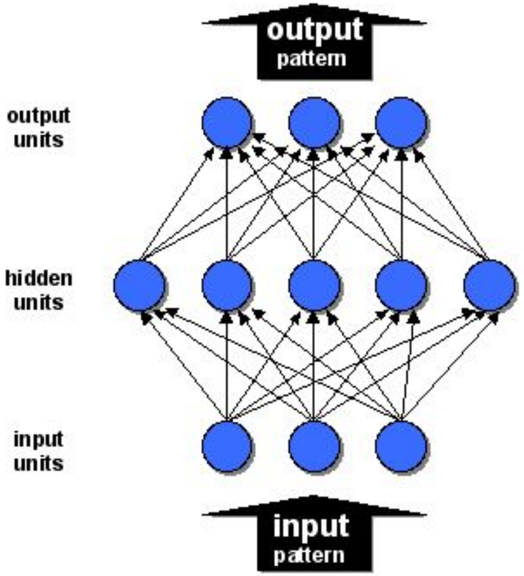
# Introduction

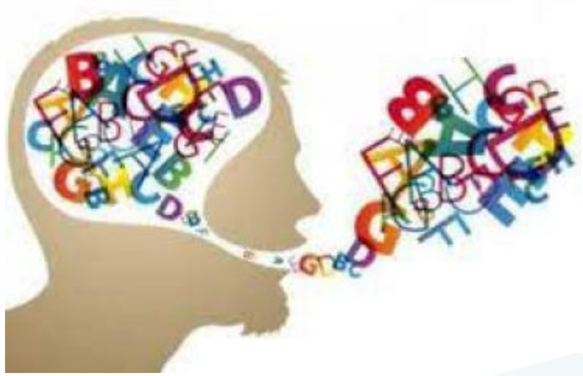
There are two main AI paradigms:

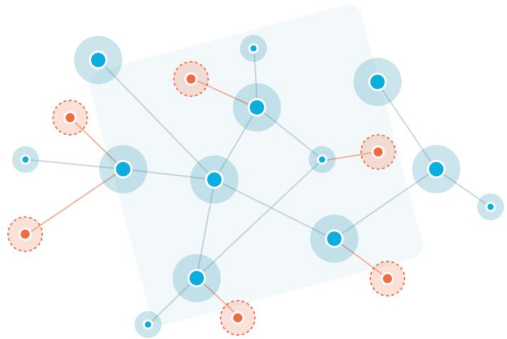
Connectionism

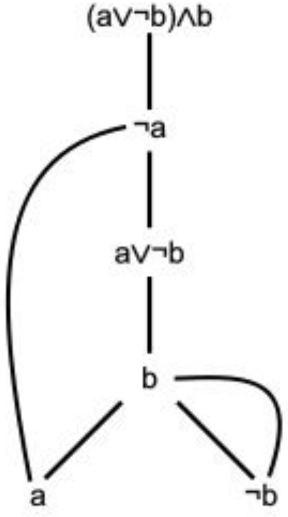





Symbolism







# Introduction

## Symbolism

- Require only a few input samples.
- Ability to explain reasoning

- Demands substantial hand-tuning and lacks true learning.
- Intolerant of ambiguous and noisy data.

## Connectionism

- Fault-Tolerant
- Generalization Capabilities

- Data Inefficient
- Lack of Comprehensibility.

# Introduction

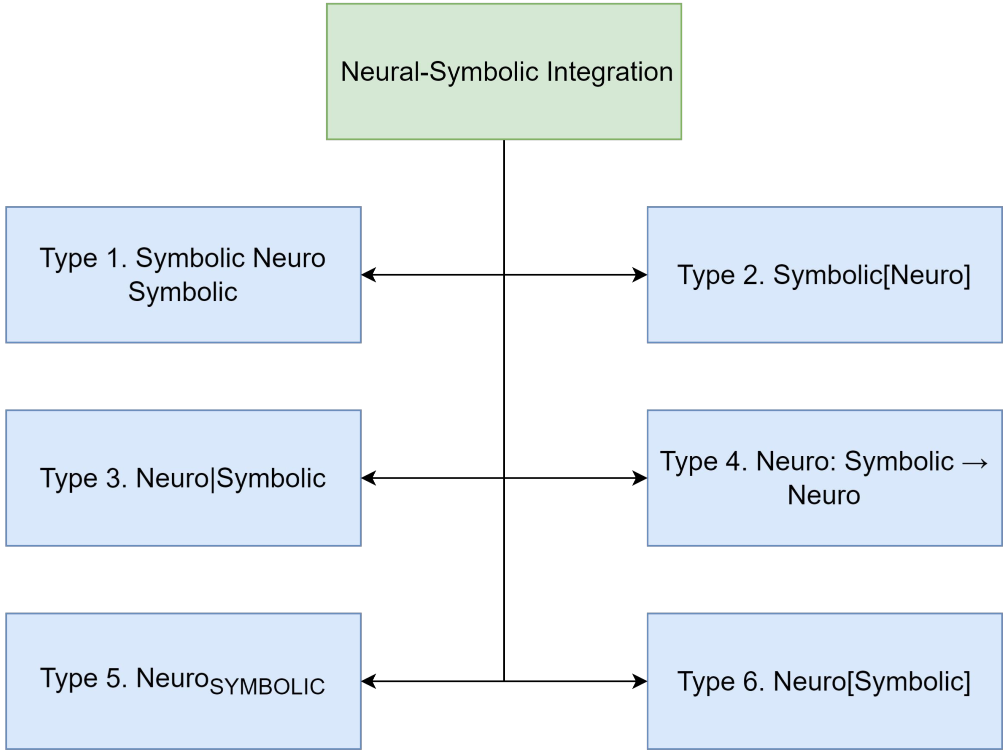
## Neural-Symbolic: Best of Both Worlds

- The two views seem to balance each other very well: why not fusing them together?
- Based on D. Kahneuman´s system 1 and system 2 theory, we can identify some parallelisms
- Connectionism can be viewed as system 1 and symbolism can be identified as system 2

# Taxonomy

Wang, W., Yang, Y., & Wu, F. (2022). Towards data-and knowledge-driven artificial intelligence: A survey on neuro-symbolic computing. *arXiv preprint arXiv:2210.15889*.

- Type 1 uses symbols as input and output via vectorization and discretization
- Type 2 and 3 use two different models (Symbolic and NN)
- Type 4 and 5 try to give DNNs "compiled" symbolic knowledge as input or feedback
- Type 6 aims to be the most faithful to human reasoning



# Type 1. Symbolic Neuro Symbolic

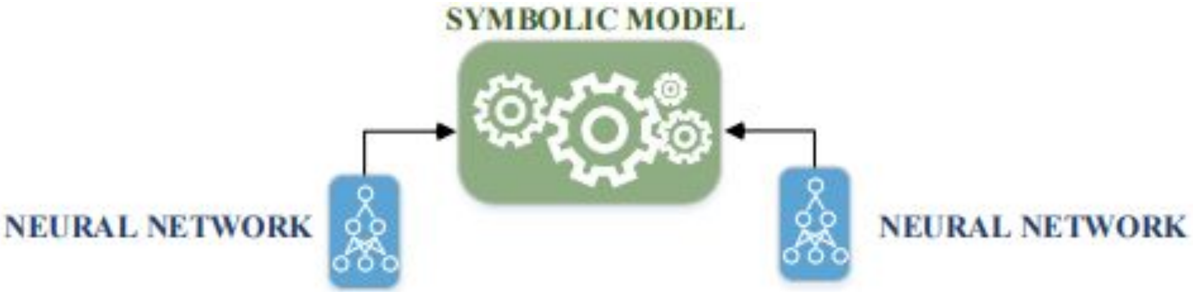
## Current Standard Operating Procedure

- Input and Output are symbols.
- Most Current NLP systems: GPT-3, word2vec...
- Possible uses: Language Translation, Graph Classification...

# Type 2. Symbolic[Neuro]

## Hybrid but overall symbolic systems

- Neural Modules are subroutines of a symbolic problem solver.
- AlphaGo: A Monte Carlo Tree Search with a neural network like the heuristic evaluation function.
- Ness: A symbolic Stack Machine with a neural network for the execution trace production.
- PLANS: A Ruled Based system with a neural perception system.



# Type 3. Neuro|Symbolic


## Hybrid but cooperative

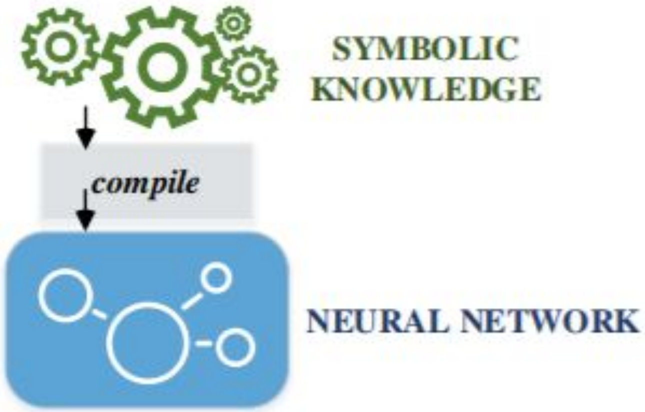
- Each part focus on different but complementary tasks.
- The interaction between both parts boost individual and collective performance.
- NS-VQA,NSPS: Program Synthesis algorithms.
- NS-CL: The symbolic module provides feedback that support gradient-based optimization of the neural module.
- PEORL: Symbolic Planning + Reinforcement Learning

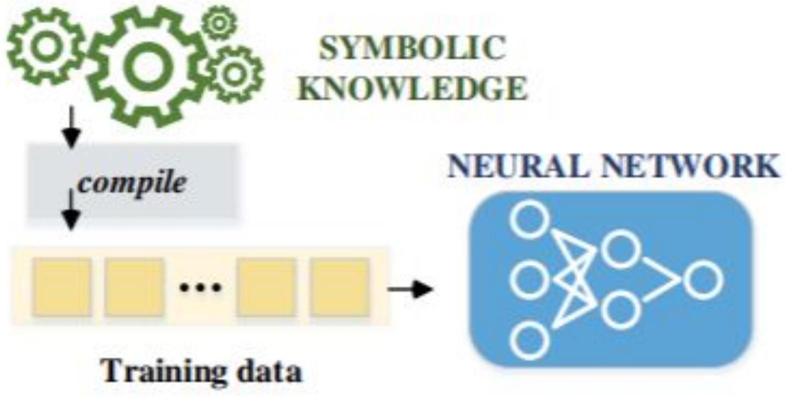
# Type 4. Neuro: Symbolic → Neuro

<u>Symbolic rules/knowledge are compiled into the architecture or training regime</u>

Main lines:

- Autoencoder representation of symbolic knowledge
- Representation of mathematical expressions as trees
- GNNs to embed entities and relations as external knowledge bases



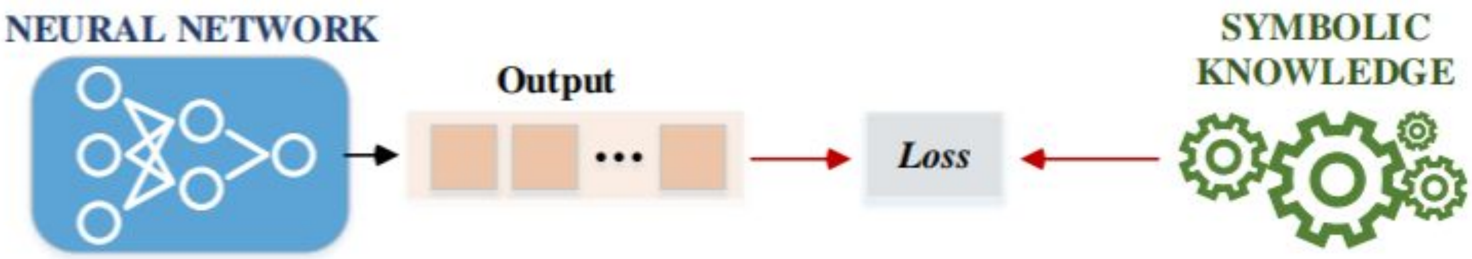



# Type 5. Neuro$_{\text{SYMBOLIC}}$

## Symbolic knowledge into soft-contraints in the loss function

Main lines:

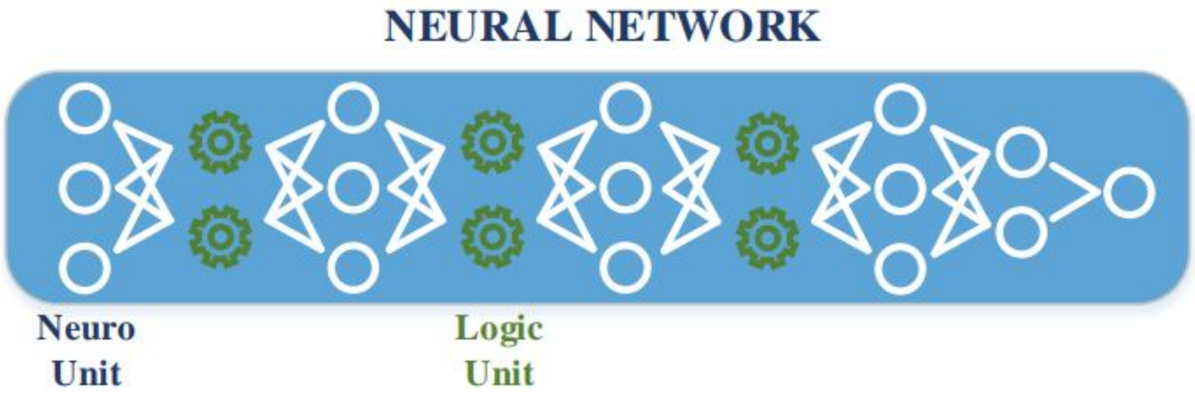
- Logic Tensor Networks (gradient based sub–symbolic learning)
  - Translate first-order logic as fuzzy relations on real numbers
- Class hierarchies (classification targets and background knowledge)



# Type 6. Neuro[Symbolic]

## Directly embed a symbolic reasoning engine inside a neural engine

- Technically capable of resembling Kahneman´s system 1 and 2
- Still very new
- Main lines:
  - Imitating logical reasoning with tensor calculus



# Open challenges

- Scalability
- Compositional Generalization
- Automated Knowledge Acquisition
- Further investigate type 6 neuro symbolic systems

# Muchas gracias

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